Aryendra Pratap Singh

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Summary

Computer Science student graduating in April 2027 who builds and deploys AI, full-stack, and database systems. Seeking full-time new grad roles in software development, AI, data, or IT.

Education

Lakehead University — Thunder Bay, ON
Honours Bachelor of Science in Computer Science | CGPA: 78.31% — Expected April 2027
Relevant Coursework: Data Structures, Algorithm Design and Analysis, Database Management Systems, Operating Systems

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript
AI and Data: Google Gemini API, Multimodal LLMs, Prompt Engineering, Computer Vision, Backtesting
Frameworks and Tools: FastAPI, REST APIs, React, Pydantic, NumPy, Streamlit, Plotly, pytest, Git, GitHub
Databases: MySQL, Schema Design, Stored Procedures, Triggers, Views, CTEs, Window Functions, Transactions
Cloud and DevOps: Docker, Docker Compose, GitHub Actions, CI, Render

Projects

RegenMed Form Checker | *Python, FastAPI, Gemini API, React* github.com/aryendrapratap/regenmed-form-checker

- Built an AI reviewer for RegenMed's challenge at the Thunder Bay AI Hackathon (September 2026) that flags missing initials, signatures, and dates on a tissue bank's scanned regulatory forms
- Reduced end-to-end review time by approximately 75% (85 to about 21 seconds) through model benchmarking, concurrent API calls with a thread-safe client, and local form identification
- Architected a hybrid pipeline: multimodal LLM extraction with the Google Gemini API, prompt engineering, and Pydantic JSON schemas, then 23 deterministic validation rules that pinpoint each error by page, row, and field
- Implemented a NumPy computer vision classifier: about 0.05 seconds per form and 34 of 34 correct confident matches
- Deployed a FastAPI REST backend and React/TypeScript frontend on Render with SHA-256 caching, HTTP 429 retry with backoff, and API keys secured in environment variables

HELOC Portfolio Analytics | *MySQL, Python, Streamlit, Docker* github.com/aryendrapratap/heloc-portfolio-analytics

- Designed a MySQL database for 300 simulated Canadian HELOC accounts over 24 months: 10-table normalized schema, 31 CHECK constraints, 10 triggers, and a transactional month-end stored procedure
- Analyzed delinquency and roll rates with 8 views and 11 SQL queries using CTEs and window functions; backtesting showed accounts with 2+ warning signs fell 30+ days behind 76.9% of the time versus 0.9%
- Automated 36 pytest tests with GitHub Actions CI on every push, and containerized the database and Streamlit/Plotly dashboard with Docker Compose for one-command setup

Aaina (GIANT JAM 2.0 Runner-Up) | *JavaScript, Phaser 3, Python, NumPy* github.com/Navyanaweli/mirror-world

- Placed runner-up (1 of 3 runner-up teams) at a 12-hour Lakehead University game jam with a 3-person team, leading game logic and level design for a mirror puzzle game on the theme "Reversal"
- Engineered a graphics-free JavaScript rules engine with fixed interfaces so 3 developers built rules, screens, and effects in parallel through Git and GitHub
- Developed a Python breadth-first search solver that found shortest solutions (9 to 18 moves) and confirmed random input cannot complete Levels 3 to 7 within 300 key presses

Experience

General Merchandise Associate (Part-Time) | Walmart Canada, Thunder Bay, ON — October 2026 – Present
Stock Unloader Associate | Walmart Canada, Thunder Bay, ON — May 2026 – Present

- Serve customers, stock merchandise accurately, and follow safety and cold-chain procedures in a high-volume store

Intern | MyGov, Delhi, India — May 2022 – July 2022

- Digitized and validated 600+ public records in structured information systems with consistent naming and metadata
- Cut manual effort by approximately 20% as part of process improvements that raised documentation accuracy
- Generated structured reports and flagged outdated or duplicate records to support decision-making

Intern | Eisenvault, Delhi, India — May 2021 – July 2021

- Validated system records, tracked updates, and identified inconsistencies in internal information systems
- Documented workflows and information-handling practices for enterprise content management and digital records